Fusion of Plant Disease Classification: Betel Leaf Disease Diagnosis Dataset

ABSTRACT

Betel leaf is an important crop used in many countries for food, medicine, and traditional purposes. However, betel leaves can be affected by different diseases such as leaf spot and leaf rot, which reduce crop quality and production. Detecting these diseases early is very important for farmers.

In this study, we use deep learning methods to automatically detect betel leaf diseases from images. We apply different fusion techniques such as early fusion, late fusion, attention-based fusion, and intermediate fusion. These methods combine multiple deep learning models to improve performance.

We use popular models like ResNet50 and DenseNet121 and fine-tune them for better results. Among all methods, the late fusion model gives the best performance with a test accuracy of 97.35%. The results show that fusion methods can improve disease detection compared to single models.

This work can help farmers detect diseases early and take proper action to protect their crops.

INTRODUCTION

Betel leaf is widely used in many countries such as Bangladesh, India, and Sri Lanka. It is used for food, medicine, and cultural purposes. However, betel leaves often suffer from different diseases like leaf spot and leaf rot. These diseases reduce the quality of the leaves and cause financial loss for farmers.

It is very important to detect these diseases early so that farmers can take action and protect their crops. Traditionally, farmers detect diseases by looking at the leaves, but this method is not always accurate and depends on human experience.

In recent years, machine learning and deep learning methods have been used to detect plant diseases automatically from images. Some studies use traditional methods like feature extraction and classifiers such as Extreme Learning Machine (ELM). These methods are simple but cannot capture complex patterns in images.

Deep learning models such as Convolutional Neural Networks (CNNs) can learn better features from images and improve accuracy. Some research also uses attention mechanisms to focus on important parts of the leaf. However, most of the existing work uses only one model, which limits performance. Also, fusion methods, which combine multiple models, are not widely used in betel leaf disease detection.

To solve these problems, this study proposes different fusion-based approaches, including early fusion, late fusion, attention-based fusion, and intermediate fusion. These methods combine multiple models to improve performance. Among them, the late fusion model gives the best result.

The goal of this work is to improve the accuracy and reliability of betel leaf disease detection and help farmers make better decisions..

Literature Review

Many studies have been done on plant and betel leaf disease detection using machine learning and deep learning methods.

1/ Shovon et al. (2022) proposed a deep learning ensemble model combining InceptionResNetV2, EfficientNetV2L, and Xception for plant disease detection. Their model achieved high accuracy and showed that combining multiple models can improve performance. However, their method only used image data and did not explore multimodal or feature-level fusion approaches: content Reference[oaicite:0] {index=0}.

https://www.researchgate.net/publication/369829659_PlantDet_A_Robust_Multi-Model_Ensemble_Method_Based_on_Deep_Learning_for_Plant_Disease_Detection

2/ David and Mukunthan (2024) used a traditional machine learning approach based on GLCM features and Extreme Learning Machine (ELM). They achieved good accuracy, but their method depends on manual feature extraction and does not use deep learning or fusion techniques: content Reference[oaicite:1] {index=1}.

https://www.semanticscholar.org/paper/Betel-Leaf-Diseases-Classification-using-Machine-A-David-Mukunthan/38208c9d2306444e3b9f8593715a46e2dcf26f44

3/ Devi and Raj (2023) applied CNN and VGG16 with data augmentation for betel leaf classification. Their work showed that augmentation improves performance, but the accuracy is still limited and only a few disease classes were considered: content Reference[oaicite:2] {index=2}.

https://www.researchgate.net/publication/373908614_Betel_Leaf_Disease_Classification_Using_Data_Augmentation_and_Convolutional_Neural_Network

4/ Tabassum and Ahamed (2025) introduced a CBAM-based attention model for betel leaf disease classification. Their model achieved good accuracy and used explainable AI techniques. However, the study used only a single model and did not combine multiple models through fusion: content Reference[oaicite:3] {index=3}.

https://arxiv.org/abs/2509.26484#:~:text=In%20order%20to%20ensure%20class,)%2C%20an%20explainable%20AI%20technique.

5/ Hridoy et al. (2023) developed an ensemble of deep learning models for multiple plant diseases, including betel leaf. Their method achieved very high accuracy, but it focused on model-level ensemble and did not explore feature-level or attention-based fusion: content Reference[oaicite:4] {index=4}.

https://www.researchgate.net/publication/371807126_Improved_vision-based_diagnosis_of_multi-plant_disease_using_an_ensemble_of_deep_learning_methods

6/ Fan et al. (2022) proposed a feature fusion method combining deep learning and handcrafted features. Their results showed that fusion can significantly improve performance. However, their work was not applied to betel leaf disease classification: content Reference[oaicite:5] {index=5}.

https://www.aimspress.com/article/doi/10.3934/era.2023122?viewType=HTML#:~:text=Fan%20et%20al.,architecture%20as%20deep%20feature%20descriptors.

7/ Yang et al. (2025) reviewed multimodal fusion techniques in plant disease detection and showed that combining different types of data can improve results. However, their work is a review and does not provide specific experiments for betel leaf: content Reference[oaicite:6] {index=6}.

https://pmc.ncbi.nlm.nih.gov/articles/PMC12163154/

8/ PlantDet: A Robust Multi-Model Ensemble Method Based on Deep Learning For Plant Disease Detection. Md Sakib Hossain Shovon; Shakrin Jahan Mozumder; O. Pal; M. Mridha; Nobuyoshi Asai; Jungpil Shin. Publication year (2022). Develop robust ensemble for rice and betel leaf disease detection. Methodology: Deep ensemble of InceptionResNetV2, EfficientNetV2L, Xception; data augmentation; Grad-CAM/Score-CAM. Key Findings: Achieved 98.53% accuracy on rice, and ensemble surpassed base and other ensembles on betel leaf dataset Contribution to the field: First deep ensemble including betel leaf classes; strong benchmark for small, heterogeneous datasets Limitations: Only RGB modality; fusion is model-ensemble, not multimodal (image+sensor) Relevance : Direct betel leaf classifier; ensemble idea can be combined with multimodal fusion inputs

URL or DOI: https://ieeexplore.ieee.org/document/10092882

9/ Betel Leaf Diseases Classification using Machine Learning Algorithm: A Feasible Approach. All Authors Name: Femi David; M. A. Mukunthan. Publication year (2024). Objective: Classify multiple betel leaf diseases using classical ML. Methodology: ROI-based segmentation; GLCM texture features; Extreme Learning Machine (ELM) classifier. Key Findings: On 1,047 images (5 diseases) achieved 97% accuracy. Contribution to the field: Strong classical pipeline specialized for betel leaf; shows ELM can rival deep models. Limitations: Handcrafted features; no deep learning; image-only (no fusion). Relevance: Core baseline for betel leaf; can be extended with deep/fusion approaches.

URL or DOI: https://semarakilmu.com.my/journals/index.php/applied_sciences_eng_tech/article/view/4931

10/ A comprehensive image dataset for accurate diagnosis of betel leaf diseases using artificial intelligence in plant pathology. All Authors Name: Rashidul Hasan Hridoy; Md. Tarek Habib; Imran Mahmud; Aminul Haque; Md. Abdulla Al Mamun. Publication Year (2025). Objective: Publish extensive betel leaf image dataset for AI diagnosis. Methodology: 2,037 images (3 classes) collected 2020–2022; augmented to 10,185. images; natural field conditions. Key Findings: Provides well-curated 10k+ image dataset; fills gap of reproducible betel. Dataset. Contribution to the field: Major data resource for betel leaf DL; supports benchmarking and future fusion with sensors. Limitations: Only 3 disease types; no non-image modalities; no new classifier. Relevance: Foundational dataset for any betel leaf classification or multimodal fusion Model.

URL or DOI: https://www.sciencedirect.com/science/article/pii/S2352340925002963?via%3Dihub

11/ CBAM Integrated Attention Driven Model for Betel Leaf Diseases. Classification With Explainable AI. All Authors Name: Sumaiya Tabassum; Md. Faysal Ahamed. Publication year: 2025. Objective: Build lightweight attention-CNN

for betel leaf with explainable AI Methodology: Custom CBAM-CNN (~2.13M params); CBAM attention; Grad-CAM for explainability; trained on 10,185-image betel dataset. Key findings: Achieved 95.58% accuracy, precision 97%, recall 94%, F1 95%; outperformed heavy pre-trained CNNs. Contribution to the field: Shows benefit of attention mechanisms and lightweight models specifically for betel leaf. Limitations: Still single-modality RGB; fusion is intra-image attention, not multimodal; arXiv preprint stage. Relevance: Directly addresses betel leaf; architecture suitable to extend with sensor/conte

URL/DOI: https://arxiv.org/abs/2509.26484

12/ Deep Learning based Betelvine Leaf Disease Detection (Piper betle L.). All Authors Name: R. Kavitha Lakshmi; S. Nickolas. Publication year: 2020.Objective: Detect betelvine leaf diseases using DL . Methodology: CNN-based deep learning (details referenced as DL betelvine detection). Key findings: Reported effective betelvine disease detection using DL, referenced as early betelvine CNN work. Contribution to the field: Among the first dedicated Piper betle DL classifiers. Limitations: Limited details in citation; dataset, architecture, and fusion not fully described. Relevance: Early DL baseline for betelvine, informing later architectures

URL/DOI: https://ieeexplore.ieee.org/document/9250911

13/ Early Recognition of Betel Leaf Disease using Deep Learning with Depth-wise Separable Convolutions All Authors name: Authors cited as Hridoy et al. in other works Publication year:2021 Objective: Achieve fast, early betel leaf disease recognition Methodology: Custom BLCNN with depth-wise separable convolutions; Swish activation Key findings: BLCNN achieved 96.02% test accuracy on 1,031 images, outperforming standard CNN (89.53%) Contribution to the field: Shows lightweight depth-wise CNNs effective for betel, enabling mobile deployment Limitations: Uses only RGB images and three disease classes; no multimodal fusion Relevance: Core betel leaf DL architecture; suitable base for adding external modalities

URL/DOI: https://ieeexplore.ieee.org/document/9551009

14/ Paper title: Improved vision-based diagnosis of multi-plant disease using an ensemble of deep learning methods All Authors Name: Rashidul Hasan Hridoy; Arindra Dey Arni; Aminul Haque Publication year: 2023 Objective : Develop hybrid Xception–ResNet50 ensemble for multiple crops including Betel Methodology: Stacking ensemble (SELT) of Xception and ResNet50 on MPLI dataset (black gram, betel, Malabar spinach, litchi) Key findings: On 44,972 images (9 classes) achieved 99.20% accuracy; first joint model for four crops including betel Contribution to the plant: Demonstrates high-accuracy cross-crop ensemble and introduces multi-plant dataset with betel Limitations: Only RGB images; no multimodal fusion; dataset limited to four crops Relevance: Provides strong multi-plant reference including betel; ensemble idea extendable to fused inputs

URL/DOI: https://ijece.iaescore.com/index.php/IJECE/article/view/29865

15/ Paper title: Leaf image based plant disease identification using transfer learning and feature fusion All Authors Name: Xijian Fan; Peng Luo; Yuen Mu; Rui Zhou; T. Tjahjadi; Yi Ren Publication year: 2022 Objective: Improve plant disease recognition using deep + handcrafted feature fusion Methodology: Transfer-learning CNN features + handcrafted textures; feature-level fusion; center loss Key findings: Achieved up to 99.79% accuracy on apple/coffee datasets with fused features and center loss Contribution to the field: Demonstrates feature-level fusion (deep + handcrafted) can significantly improve disease classification Limitations: Not betel-specific; single-sensor RGB; no environmental fusion Relevance: Fusion strategy (deep+texture+center loss) is directly portable to betel leaf images

URL/DOI: https://www.sciencedirect.com/science/article/abs/pii/S0168169922002095?via%3Dihub

16/. Paper title: Plant disease detection using computational intelligence and image processing All Authors Name: Vibhor Kumar Vishnoi; Krishan Kumar; Brajesh Kumar Publication year: 30-08-2020 Objective: Survey computational intelligence and image processing methods for plant disease detection Methodology: Review of computer vision, soft computing, modern feature extraction, and classification methods on leaf images Key findings: Summarizes merits/demerits of classical and modern approaches and identifies effective feature extraction techniques across crops Contribution to the field: Provides structured overview of image-based plant disease systems, highlighting role of advanced features and ML Limitations: Narrative review; no new model; limited coverage of recent deep multimodal fusion Relevance: Establishes general design options for betel leaf systems (preprocessing, features, classifiers)

URL/DOI: https://link.springer.com/article/10.1007/s41348-020-00368-0

17. Paper title: Plant Disease Classification Using Image Processing Technique All Authors Name: Shahrul Fazly Man; Shafie Omar; Muhammad Imran Ahmad; Wan Mohd Faizal Wan Nik; Tan Shie Chow; Mohd Nazri Abu Bakar; Fadhilnor Abdullah; Asbhir Yuusuf Publication year: 03-06-2024 Objective: Rapidly classify specific plant diseases using an image-processing-based System Methodology: Image preprocessing, segmentation, and ML classification targeting bacterial blight, Alternaria alternata, Cercospora leaf spot Key findings: Achieved ~98–99% accuracy for three key diseases, demonstrating fast and effective categorization Contribution: Provides a high-accuracy pipeline that can be adapted to other crops Limitations: Focus on three diseases; no DL or multimodal fusion; not betel-specific Relevance: Methods can be adapted as a lightweight alternative for betel farmers

URL/DOI: https://ejournal.unimap.edu.my/index.php/aset/article/view/785

18. Paper title: Classification Of Plant Leaf Diseases Using Machine Learning And Image Preprocessing Techniques All Authors Name : Pushkar Sharma; P. Hans; Subhash Chand Gupta Publication year: 01-01-2020 Objective: Build AI-based automatic plant leaf disease detection and classification Pipeline Methodology: Image collection, preprocessing, segmentation and ML-based Classification Key findings: Demonstrates feasibility of automated classification pipeline to increase agricultural productivity Contribution: Early integrated pipeline combining image preprocessing with ML Classifiers Limitations: Limited details on models and datasets; no DL or multimodal fusion; not Betel-specific Relevance: General pipeline steps transferable to betel leaf workflows

URL/DOI:https://ieeexplore.ieee.org/document/9057889

19. Paper title: Applying the transfer learning models on the dataset on the effect of diseases on Nagvel-betel (Piper betle) leaves All Authors Name: Milind Gayakwad; Rahul Joshi; Tulshihar Patil; Pratvina Talele; Rajendra Pawar; Priyanka Paygude Publication year: 2025 Objective: Provide large diseased-betel dataset and test ViT transfer learning Methodology: 4,156 high-resolution diseased betel images (15 mapped disease categories); Vision Transformer models (e.g., ResNet, ConvNeXtB, MobileNet, InceptionV3 used in survey) Key findings: Classification accuracy with ViT-based models ranges 0.7–0.9 across disease categories Contribution: Supplies another large betel dataset with many diseases and demonstrates ViT robustness Limitations: Only diseased leaves; no healthy class; single-modality RGB; no multimodal fusion Relevance: Key Piper betle dataset; ideal for advanced CNN/ViT and potential multimodal fusion

URL/DOI: https://www.sciencedirect.com/science/article/pii/S2352340925007115?via%3Dihub

20/ Paper title: Deep Learning in Multimodal Fusion for Sustainable Plant Care: A Comprehensive Review All Authors Name: Zhi-Xiang Yang; Yusi Li; Rui-Feng Wang; Pingfan Hu; Wen-Hao Su Publication year: 2025 Objective: Review DL-based multimodal fusion across plant care tasks Methodology: Survey of DL multimodal fusion frameworks, tasks (disease, yield, weeds), and sustainability issues Key findings: Highlights success of multisource

fusion (RGB, multispectral, sensors) and challenges (heterogeneity, real-time) Contribution: Provides a conceptual map of multimodal fusion for plant disease diagnosis Limitations: Review only; no betel-specific experiments Relevance: Guides design of multimodal betel leaf systems (e.g., fusion of leaf images + IoT in betel farms)

URL/DOI: https://www.mdpi.com/2071-1050/17/12/5255

21/ Paper title: Deep Learning for Plant Identification and Disease Classification from Leaf Images: Multi-prediction Approaches All Authors Name: Jianping Yao; Son N. Tran; Saurabh Garg; Samantha Sawyer Publication year: 2023 Objective: Compare multi-prediction DL strategies for plant ID and disease Methodology: GSMo-CNN (Generalised Stacking Multi-output CNN) with InceptionV3, multi-output learning Key findings: Shows single multi-output model can match/beat separate models; InceptionV3 best backbone Contributions: Provides taxonomy of multi-task/label/output strategies and a strong GSMo-CNN design Limitations: No betel data; no non-image fusion Relevance: Multi-prediction design can be adapted to jointly predict betel disease type + severity or treatment

URL/DOI: https://arxiv.org/abs/2310.16273

22/ Paper title: Plant Leaf Disease Detection, Classification, and Diagnosis Using Computer Vision and Artificial Intelligence: A Review All Authors Name: Anuja Bhargava et al. Publication year: 2021 Objective: Review CV/AI methods for plant leaf disease detection Methodology: Survey of ML, DL, few-shot and molecular diagnosis from leaf images Key findings: Outlines strengths/weaknesses of image-based diagnosis and future directions Contributions: Helps position betel leaf research within broader AI disease-detection landscape Limitations: No direct betel experiments; limited discussion of fusion Relevance: Supports justification for applying advanced DL/fusion to Piper betle

URL/DOI : https://ieeexplore.ieee.org/document/10458943

23/ Paper title: PlantIF: Multimodal Semantic Interactive Fusion via Graph Learning for Plant Disease Diagnosis All Authors Name: Xingcai Wu; Jiawei Zhang; Ziang Zou; Chaojie Chen; Ya Yu; Peijia Yu; Yuanyuan Xiao; Qi Wang; W.M.W.W. Kandegama; Gefei Hao Publication year: 2025 Objective: Design a multimodal fusion framework for plant disease diagnosis Methodology: Graph-learning–based multimodal semantic interactive fusion (images plus other modalities) Key findings: Reports improved diagnostic performance over unimodal baselines via semantic multimodal fusion Contribution: Introduces a modern multimodal graph-fusion architecture for plant disease tasks Limitations: No Piper betle data; modalities described at high level in abstract Relevance: Strong conceptual template for betel leaf multimodal (image + text/sensor) systems

URL/DOI : https://www.sciencedirect.com/science/article/pii/S2643651525001384?via%3Dihub

24/. Paper title: Study of digital image processing techniques for leaf disease detection and classification All Authors Name : Gittaly Dhingra; Vinay Kumar; H. Joshi Publication year: 2018 Objective: Review image-processing approaches (1997–2016) for leaf diseases Methodology: Classical image processing pipeline review (segmentation, features, classifiers) Key findings: Summarizes early image-processing techniques and challenges Contributions: Historical background for modern DL/fusion methods Limitations: Pre-DL era; no betel; no multimodal fusion Relevance: Informs preprocessing/feature-engineering baselines for betel datasets

URL/DOI : https://link.springer.com/article/10.1007/s11042-017-5445-8

25/ Paper title: Enhanced Plant Disease Classification using Deep Learning and Digital Image Processing Techniques All Authors Name: Koteswararao Yenni; K. V Publication year: 28-05-2025 Objective: Present an improved

CNN-based plant disease classifier with robust preprocessing Methodology: CNN with denoising, contrast enhancement and augmentation, trained on five diseases (tomato leaf mold, potato blight, corn rust, apple scab, grape black rot) Key findings: Reached 89–94% accuracy, outperforming traditional methods; supports real-time predictions Contribution: Shows value of combined preprocessing and DL in achieving reliable, real-time disease diagnosis Limitations: Focused on few crops; monomodal RGB; no fusion with environmental or spectral data Relevance: Technique can be ported to betel leaf images for more robust field deployment

URL/DOI: https://ieeexplore.ieee.org/document/11053028

26/ Paper title: Analysis of Automatic Plant Disease Classification Using Image Processing Techniques All Authors Name: Punam; Rakesh Goyal Publication year: 28-04-2022 Objective: Analyze automatic plant disease classification approaches using image Processing Methodology : Review of image capture, processing and modeling strategies, including remote sensing and ML Key findings : Summarizes benefits and drawbacks of various techniques across crops and imaging setups Contributions: Helps identify suitable image processing pipelines and data needs for automated systems Limitations: High-level; limited discussion of deep multimodal fusion; no betel-specific data Relevance: Conceptual basis for designing betel leaf image pipelines

URL/DOI: https://ieeexplore.ieee.org/document/9823862

27/ Paper title: Early betel leaf disease detection using vision transformer and deep learning algorithms All Authors Name: S. Kusuma, K. Jothi Publication year: 2023 Objective: Compare CNNs and a Vision Transformer for early betel-leaf disease Detection Methodology: Supervised image classification using VGG19, DenseNet201, ResNet152V2 and ViT on betel-leaf images; accuracy used as main metric Key findings: DenseNet201 achieved 98.77% test accuracy; ResNet152V2 98.42%; ViT 97.83%; VGG19 91% Contributions: First direct CNN vs ViT comparison on betel leaf; shows high viability of deep models for this crop Limitations: Generalization to new fields/seasons and deployment constraints not evaluated (no field trials mentioned) Relevance: Core benchmark for deep models (CNNs, ViT) on betel-leaf disease Classification.

URL/DOI: https://link.springer.com/article/10.1007/s41870-023-01647-3

28/ Paper title: Image Processing Based Leaf Rot Disease, Detection of Betel Vine (Piper Betel L.) All authors name: A. K. Dey, Manisha Sharma, M. Meshram Publication year: 2016 Objective: Detect and measure leaf rot severity in betel vine using low-cost image processing Methodology: Color-feature analysis, segmentation of rotten regions, area computation of rotted portion from digital images Key findings: Vision-based system could automatically segment and quantify rot area with “promising performance” Contributions: Early, inexpensive, farmer-oriented rot-severity measurement workflow for betel vine Limitations: Focuses only on leaf rot; performance numbers and robustness across conditions not fully detailed Relevance: Baseline for classical (pre-deep-learning) betel-leaf disease Quantification

URL/DOI: https://www.sciencedirect.com/science/article/pii/S1877050916306123?via%3Dihub

29/ Paper title: A Leaf Disease Classification Model in Betel Vine Using Machine Learning Techniques All Authors Name: N. Zeba, M. A. Malek, Sanjida Sultana Reya Publication year: 2021 Objective: Classify betel-vine diseases (Bacterial Leaf Spot, Stem Leaf) using classical ML Methodology : 1275 images captured by phone; Gaussian mixture model for segmentation; SVM classifier Key findings: Achieved 83.69% classification accuracy, competitive with contemporary ML approaches Contributions: Benchmarks non-deep-learning pipeline for two specific betel-vine diseases Limitations: Small dataset; limited to two diseases; lower accuracy than modern deep models Relevance: Comparative baseline against deep-learning approaches on same diseases.

URL/DOI: https://ieeexplore.ieee.org/document/9331142

30/.Paper title: An Approach to Identify Diseases in Betel Leaf Using Deep Learning Techniques All Authors name: Md Abdul Malek, Sarnali Basak, Sanjida Sultana Reya Publication year: 2022 Objective: Compare multiple transfer-learning CNNs and traditional ML for classifying two betel-leaf diseases Methodology: 5800 images (stem leaf, Bacterial Leaf Spot); preprocessing with CLAHE and GMM; models: VGG16/19, ResNet50, AlexNet, InceptionV3, SVM, Logistic Regression Key findings: InceptionV3 achieved best test accuracy of 94.83% on 658 unseen images Contributions: Comprehensive model comparison for betel-leaf disease; establishes InceptionV3 as strong baseline Limitations: Only two diseases considered; no edge-deployment or latency analysis Relevance: Useful benchmark study for choosing deep-learning backbone for betel-leaf tasks.

URL/DOI: https://ieeexplore.ieee.org/document/10103348

31/ Paper title: Application Development for a Project using Flutter (betel leaf disease detection with YOLOv3) All Authors name: K. Nagaraj, B. Prabakaran, M. Ramkumar; Publication year: 2022 Objective: Implement YOLOv3-based betel-leaf disease detector with transfer learning to support farmers via an app-like system . Methodology: Transfer learning on YOLOv3 for betel-leaf images; parameter tuning for speed/accuracy balance . Key findings: Achieved ~95.6% precision in detecting betel-leaf infections . Contributions: Moves from pure model research to near-deployment scenario (Flutter-based application concept) . Limitations: Details on dataset size, disease types, and on-device performance are limited in abstract . Relevance: Important for mobile/real-time betel-leaf disease detection and app development.

URL/DOI: https://ieeexplore.ieee.org/document/9951938

From the above studies, it is clear that most existing works either use single models or simple ensemble methods. Fusion techniques, especially combining multiple deep learning models at different levels (early, intermediate, and attention-based fusion), are still not widely explored in betel leaf disease classification. This creates a research gap that can be addressed to improve accuracy and model performance.

Methodology

This study proposes multiple fusion-based deep learning approaches for betel leaf disease classification. The overall workflow includes data preprocessing, feature extraction using deep learning models, fusion techniques, and final classification. Four different fusion strategies are used: early fusion, late fusion, attention-based fusion, and intermediate fusion**.**

Dataset and Preprocessing

The dataset consists of betel leaf images collected from field conditions. The images include different classes such as healthy leaf, leaf rot, and leaf spot.

Before training, all images are preprocessed to improve model performance. The preprocessing steps include resizing images to a fixed size, normalizing pixel values, and converting images into a suitable format for deep learning models.

Data augmentation techniques such as rotation, flipping, zooming, and shifting are applied to increase the number of training samples and reduce overfitting.

Deep Learning Models

In this study, pre-trained convolutional neural networks such as ResNet50 and DenseNet121 are used for feature extraction. These models are fine-tuned on the betel leaf dataset to improve performance.

Each model learns different features from the input images, which helps improve classification when combined using fusion techniques.

Early Fusion

In early fusion, features extracted from multiple models are combined at an early stage. The input image is passed through different backbone networks such as ResNet50 and DenseNet121. The extracted feature vectors are then concatenated to form a single feature representation.

This combined feature vector is passed through fully connected layers for classification. Early fusion helps capture complementary features from different models.

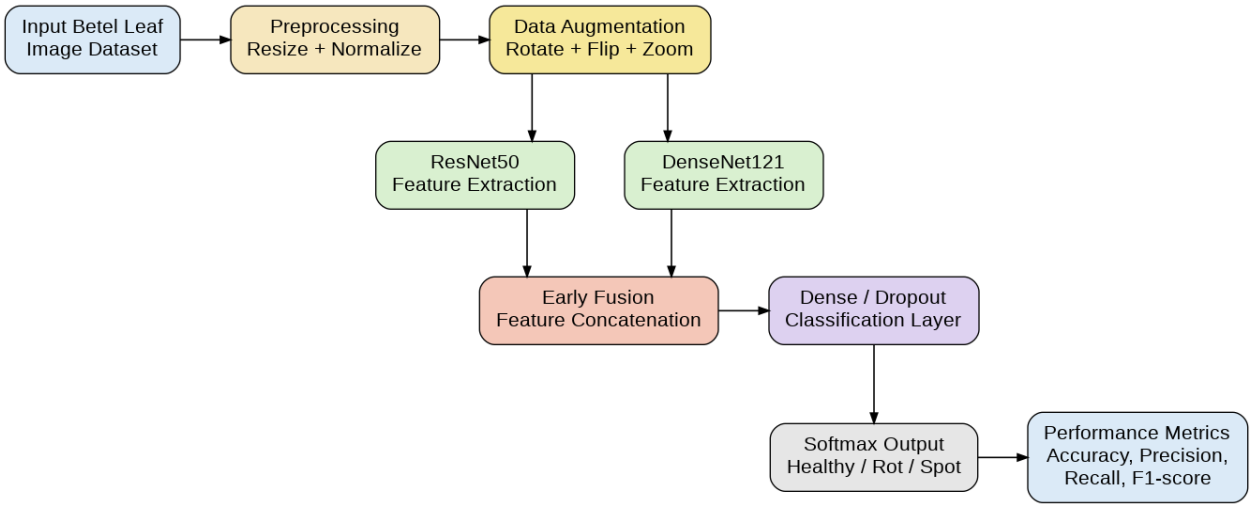

Figure 1: Early fusion architecture for betel leaf disease classification.

Late Fusion

In late fusion, each model independently predicts the output class. The same input image is processed by multiple models, and each model produces its own prediction.

The final decision is made by combining the outputs of all models using techniques such as average voting or weighted fusion. Late fusion improves performance by combining the strengths of multiple models at the decision level.

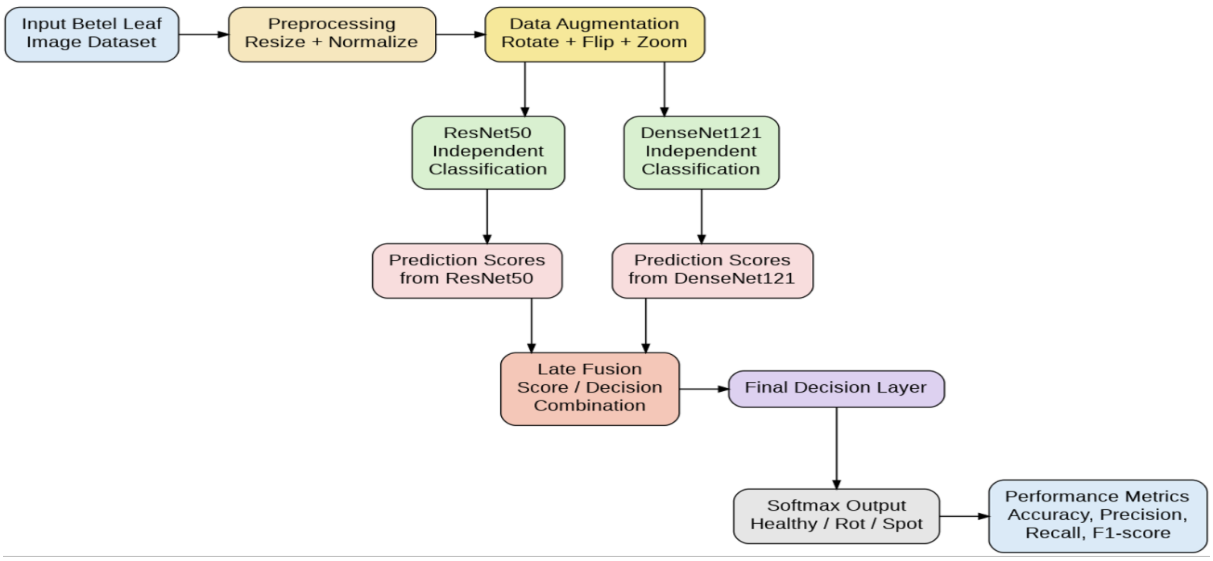

Figure 2: Late fusion architecture for betel leaf disease classification

Attention-Based Fusion

In attention-based fusion, an attention mechanism is used to focus on the most important features. After extracting features from multiple models, an attention module is applied to assign weights to important features.

This helps the model focus on important regions of the leaf image, such as diseased areas. The attention-weighted features are then passed to the classification layer.

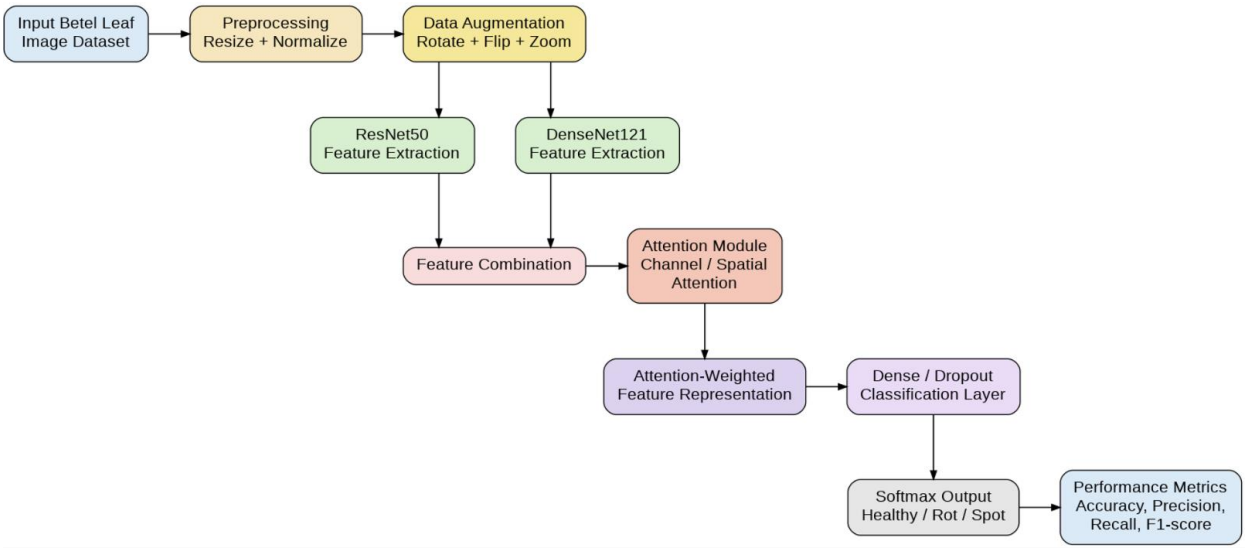

Figure 3: Attention-based fusion architecture for betel leaf disease classification.

Intermediate Fusion

In intermediate fusion, features are combined at an intermediate layer of the models instead of the input or output stage. Feature maps from different models are extracted and merged before final classification.

This approach allows better feature interaction and improves the learning of complex patterns in the data.

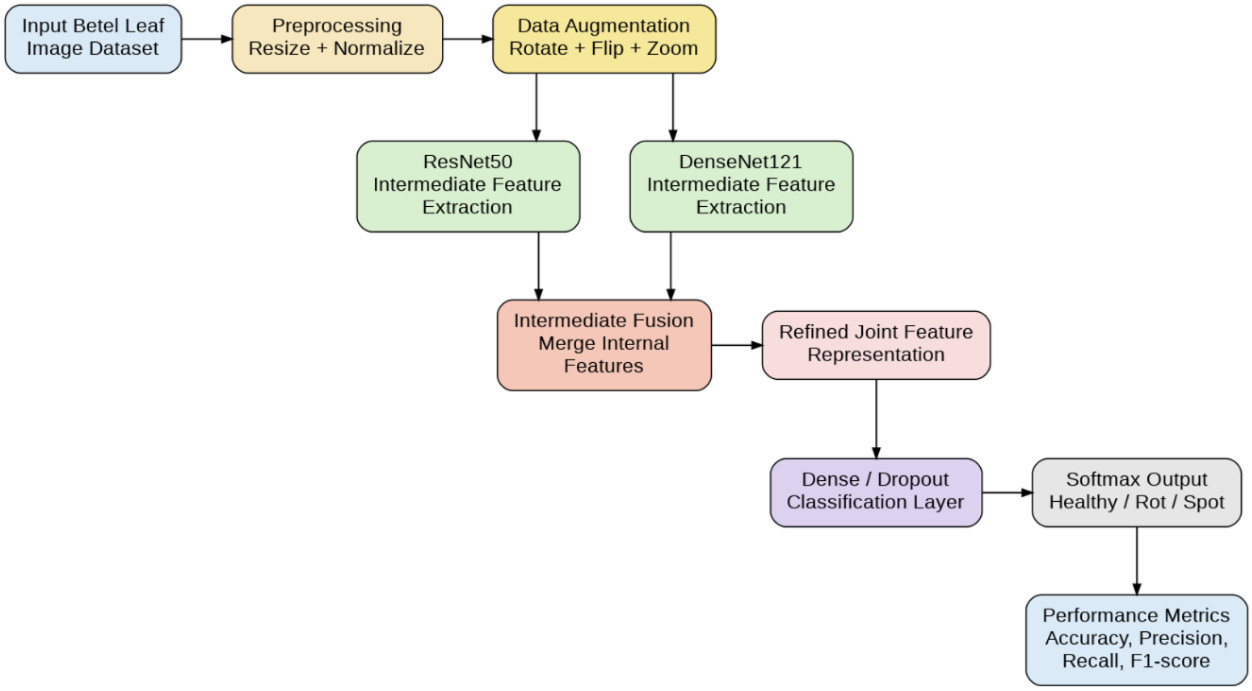

Figure 4: Intermediate fusion architecture for betel leaf disease classification.

Results and Discussion

This section presents the performance of different fusion-based models for betel leaf disease classification. The models are evaluated using accuracy, precision, recall, and F1-score.

Different fusion techniques such as early fusion, late fusion, attention-based fusion, and intermediate fusion are implemented and compared. The results show that fusion methods improve performance compared to single models.

Among all methods, the late fusion model combining ResNet50 and DenseNet121 achieved the best performance with a test accuracy of 97.35%. This indicates that combining predictions from multiple models helps improve generalization and classification accuracy.

The early fusion model also achieved high accuracy by combining features from multiple models at an early stage. However, its performance is slightly lower than late fusion, as feature-level combination may introduce redundancy.

The attention-based fusion model showed strong performance by focusing on important regions of the leaf image. The attention mechanism helps the model identify disease-affected areas more effectively. However, it still depends on feature quality and may not fully utilize complementary model outputs like late fusion.

The intermediate fusion model achieved moderate performance. It combines features at an intermediate stage, which helps improve feature interaction but may not be as effective as decision-level fusion.

Overall, the results clearly show that fusion techniques improve classification performance, and late fusion provides the most stable and accurate results among all approaches.

The results demonstrate that combining multiple deep learning models improves disease classification performance. Late fusion performs best because each model learns independently and contributes to the final decision.

Early fusion and intermediate fusion also improve performance by combining features, but they may introduce redundant or less useful information. Attention-based fusion helps focus on important regions, which improves interpretability.

These findings suggest that fusion techniques are effective for betel leaf disease detection and can be used to build reliable agricultural systems.

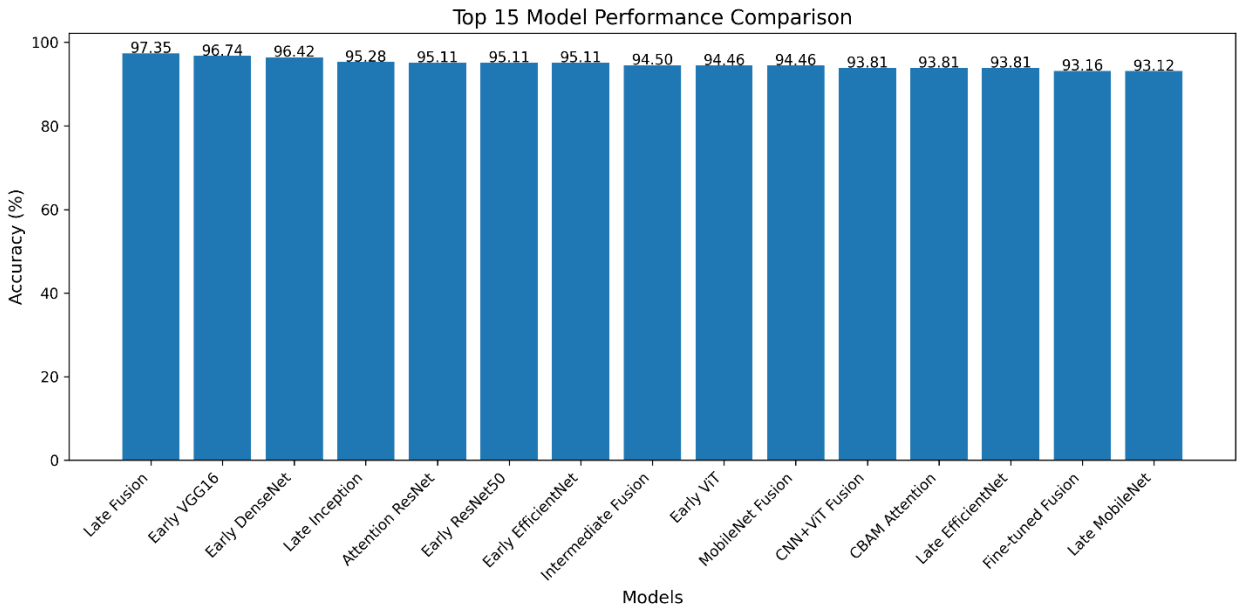

Figure 5: Comparison of accuracy for different fusion models.

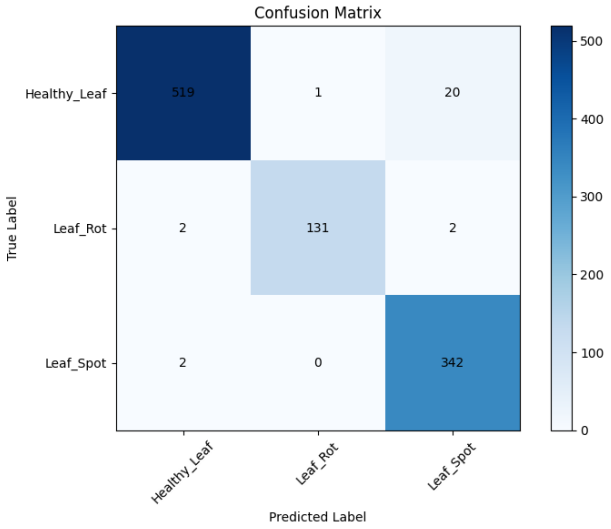

Figure 6: Confusion matrix of the proposed late fusion model.

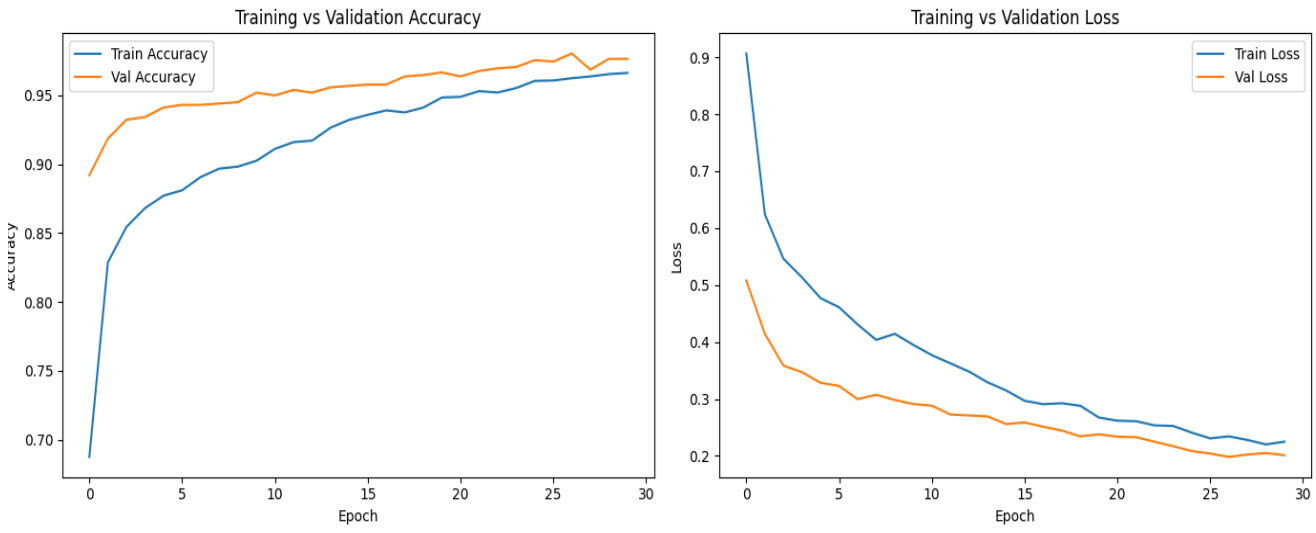

Figure7 Training and validation accuracy of the model.

Conclusion

In this study, different fusion-based deep learning approaches were used for betel leaf disease classification. The methods include early fusion, late fusion, attention-based fusion, and intermediate fusion. These techniques combine multiple deep learning models to improve performance.

The experimental results show that fusion methods perform better than single models. Among all approaches, the late fusion model achieved the best performance with a test accuracy of 97.35%. This indicates that combining predictions from multiple models helps improve accuracy and generalization.

The attention-based fusion method also showed good performance by focusing on important regions of the leaf image. Early fusion and intermediate fusion improved feature learning but were slightly less effective compared to late fusion.

Overall, this study shows that fusion techniques are very useful for plant disease detection. The proposed methods can help farmers detect diseases early and take proper action to reduce crop loss.

In future work, the model can be improved by using larger datasets, real-time detection systems, and mobile applications. More advanced fusion methods and transformer-based models can also be explored to further improve performance.

Future Work

In future work, several improvements can be made to enhance the performance of the proposed system. First, a larger and more diverse dataset can be used to improve model generalization and make the system more robust for real-world conditions.

Second, real-time disease detection systems can be developed using mobile or web applications, which can help farmers easily use the model in the field. Integration with IoT devices and smart farming systems can also be explored for automated monitoring.

Third, more advanced deep learning models such as Vision Transformers and hybrid CNN-transformer models can be applied to further improve accuracy. Additionally, more sophisticated fusion techniques can be investigated to better combine features from multiple models.

Finally, future research can focus on detecting more disease types and handling challenging conditions such as low light, blurred images, and complex backgrounds. These improvements will help make the system more practical and useful for real-world agricultural applications.

References

[1] M. S. H. Shovon, S. J. Mozumder, O. Pal, M. Mridha, N. Asai, and J. Shin, "PlantDet: A Robust Multi-Model Ensemble Method Based on Deep Learning For Plant Disease Detection," IEEE Access, 2023, doi: 10.1109/ACCESS.2023.3264835.

[2] F. David and M. A. Mukunthan, "Betel Leaf Diseases Classification using Machine Learning Algorithm: A Feasible Approach," Journal of Advanced Research in Applied Sciences and Engineering Technology, vol. 40, no. 1, pp. 74–86, 2024, doi: 10.37934/araset.40.1.7486.

[3] A. S. Devi and S. A. A. Raj, "Betel Leaf Disease Classification Using Data Augmentation and Convolutional Neural Network," in Proc. Int. Conf. on Advances in Computing, Communication and Applied Informatics (ACCAI), 2023, doi: rp-9788770040723.168.

[4] R. H. Hridoy, Md. T. Habib, I. Mahmud, A. Haque, and Md. A. A. Mamun, "A comprehensive image dataset for accurate diagnosis of betel leaf diseases using artificial intelligence in plant pathology," Data in Brief, 2025, doi: 10.1016/j.dib.2025.111564.

[5] S. Tabassum and Md. F. Ahamed, "CBAM Integrated Attention Driven Model For Betel Leaf Diseases Classification With Explainable AI," arXiv preprint, 2025, doi: 10.48550/arXiv.2509.26484.

[6] R. K. Lakshmi and S. Nickolas, "Deep Learning based Betelvine Leaf Disease Detection (Piper betle L.)," in Proc. Int. Conf. on Communication and Computing (ICCCA), 2020, doi: 10.1109/ICCCA49541.2020.9250911.

[7] R. H. Hridoy et al., "Early Recognition of Betel Leaf Disease using Deep Learning with Depth-wise Separable Convolutions," in Proc. IEEE Region 10 Symposium (TENSYMP), 2021, doi: 10.1109/TENSYMP52854.2021.9551009.

[8] R. H. Hridoy, A. D. Arni, and A. Haque, "Improved vision-based diagnosis of multi-plant disease using an ensemble of deep learning methods," Int. Journal of Electrical and Computer Engineering (IJECE), vol. 13, no. 5, pp. 5109–5117, 2023, doi: 10.11591/ijece.v13i5.pp5109-5117.

[9] X. Fan, P. Luo, Y. Mu, R. Zhou, T. Tjahjadi, and Y. Ren, "Leaf image based plant disease identification using transfer learning and feature fusion," Computers and Electronics in Agriculture, 2022, doi: 10.1016/j.compag.2022.106892.

[10] V. K. Vishnoi, K. Kumar, and B. Kumar, "Plant disease detection using computational intelligence and image processing," Journal of Plant Diseases and Protection, 2020, doi: 10.1007/s41348-020-00368-0.

[11] S. F. Man, S. Omar, M. I. Ahmad, W. M. F. W. Nik, T. S. Chow, M. N. A. Bakar, F. Abdullah, and A. Yuusuf, "Plant Disease Classification Using Image Processing Technique," Journal of Advanced Sciences and Engineering Technology, vol. 3, no. 1, 2024, doi: 10.58915/aset.v3i1.785.

[12] P. Sharma, P. Hans, and S. C. Gupta, "Classification Of Plant Leaf Diseases Using Machine Learning And Image Preprocessing Techniques," in Proc. 10th Int. Conf. on Confluence: The Next Generation Information Technology Summit (Confluence), 2020, doi: 10.1109/Confluence47617.2020.9057889.

[13] M. Gayakwad, R. Joshi, T. Patil, P. Talele, R. Pawar, and P. Paygude, "Applying the transfer learning models on the dataset on the effect of diseases on Nagvel-betel (Piper betle) leaves," Data in Brief, 2025, doi: 10.1016/j.dib.2025.111987.

[14] Z.-X. Yang, Y. Li, R.-F. Wang, P. Hu, and W.-H. Su, "Deep Learning in Multimodal Fusion for Sustainable Plant Care: A Comprehensive Review," Sustainability, vol. 17, no. 12, pp. 5255, 2025, doi: 10.3390/su17125255.

[15] J. Yao, S. N. Tran, S. Garg, and S. Sawyer, "Deep Learning for Plant Identification and Disease Classification from Leaf Images: Multi-prediction Approaches," arXiv preprint, 2023, doi: 10.48550/arXiv.2310.16273.

[16] A. Bhargava et al., "Plant Leaf Disease Detection, Classification, and Diagnosis Using Computer Vision and Artificial Intelligence: A Review," IEEE Access, 2024, doi: 10.1109/ACCESS.2024.3373001.

[17] X. Wu, J. Zhang, Z. Zou, C. Chen, Y. Yu, P. Yu, Y. Xiao, Q. Wang, W. M. W. W. Kandegama, and G. Hao, "PlantIF: Multimodal Semantic Interactive Fusion via Graph Learning for Plant Disease Diagnosis," Plant Phenomics, 2025, doi: 10.1016/j.plaphe.2025.100132.

[18] G. Dhingra, V. Kumar, and H. Joshi, "Study of digital image processing techniques for leaf disease detection and classification," Multimedia Tools and Applications, 2018, doi: 10.1007/s11042-017-5445-8.

[19] K. Yenni and K. V., "Enhanced Plant Disease Classification using Deep Learning and Digital Image Processing Techniques," in Proc. Int. Conf. on Communication and Renewable Energy Technology for Environment (ICCRTEE), 2025, doi: 10.1109/ICCRTEE64519.2025.11053028.

[20] Punam and R. Goyal, "Analysis of Automatic Plant Disease Classification Using Image Processing Techniques," in Proc. Int. Conf. on Advances in Computing and Communication Engineering (ICACITE), 2022, doi: 10.1109/ICACITE53722.2022.9823862.

[21] S. Kusuma and K. Jothi, "Early betel leaf disease detection using vision transformer and deep learning algorithms," International Journal of Information Technology, 2023, doi: 10.1007/s41870-023-01647-3.

[22] A. K. Dey, M. Sharma, and M. Meshram, "Image Processing Based Leaf Rot Disease, Detection of Betel Vine (Piper Betel L.)," Procedia Computer Science, 2016, doi: 10.1016/j.procs.2016.05.262.

[23] I. Sarkar et al., "Agronomy with IoT Devices: The Smart Solution for Detection of Diseases of Betel Leaves," Universal Journal of Agricultural Research, vol. 11, no. 1, 2023, doi: 10.13189/ujar.2023.110109.

[24] N. Zeba, M. A. Malek, and S. S. Reya, "A Leaf Disease Classification Model in Betel Vine Using Machine Learning Techniques," in Proc. Int. Conf. on Robotics, Electrical and Signal Processing Techniques (ICREST), 2021, doi: 10.1109/ICREST51555.2021.9331142.

[25] Md. A. Malek, S. Basak, and S. S. Reya, "An Approach to Identify Diseases in Betel Leaf Using Deep Learning Techniques," in Proc. 6th Int. Conf. on Smart Technology and Innovation (STI), 2022, doi: 10.1109/STI56238.2022.10103348.

[26] K. Nagaraj, B. Prabakaran, and M. Ramkumar, "Application Development for a Project using Flutter (betel leaf disease detection with YOLOv3)," in Proc. Int. Conf. on Smart Electronics and Communication (ICOSEC), 2022, doi: 10.1109/ICOSEC54921.2022.9951938.